

are outside scope. No expert gold comparability labels, paper-level locator extraction, generated recommendation evaluation, or laboratory replication is claimed. Future work must calibrate detector profiles on held-out agent traces, group splits by DOI and device family, add expert adjudication, test decentralized execution, and test at least one second adapter.

Scientific responsibility and broader impact. Protocol safety is conditional on adapter soundness. An incomplete domain rule can admit wrong evidence, and excessive abstention can disproportionately suppress poorly reported subfields. EviCoord therefore records reasons for rejection, exposes unresolved uncertainty, escalates unresolved severe conflicts on high-impact tasks to a human scientist, and abstains on lower-impact cases. It cannot replace replication, safety assessment, authorship responsibility, or institutional review. Data release must preserve source licenses and avoid distributing copyrighted full text.

AI-use disclosure. An OpenAI Codex assistant was used to synthesize project notes, assist with analysis and visualization code, search candidate literature, and edit the manuscript. All numerical claims were generated by the disclosed reproducibility scripts; no model or expert experiment was invented. The human authors remain responsible for research design, source verification, code, results, interpretation, and final submission.

8 CONCLUSION

EviCoord establishes evidence-state coordination as a general-purpose foundation for multi-agent AI4S. Typed Evidence Envelopes, a reusable Core–Adapter separation, provenance and comparability gates, conflict isolation, verifier diversity, and bounded terminal actions turn scientific evidence into a traceable, verifiable, and governable coordination object. The resulting architecture provides a clear path to cross-domain deployment: the coordination core remains stable, while expert knowledge enters through explicit, field-specific adapters. In the real-record-grounded perovskite instantiation, controlled fault experiments demonstrate that adaptive evidence-state routing substantially reduces unsafe output while preserving useful coverage and controlling verification cost. Together, the formal guarantees, implementation audits, and empirical results position EviCoord as a practical, extensible coordination paradigm for scaling reliable multi-agent scientific discovery across disciplines.

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